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#!/bin/bash

# =====
# Pairwise GWAS (PW-GWAS)
#
# Example analysis comparing two GWAS phenotypes.
#
# Requirements:
#   - gwas-pw
#   - Harmonised GWAS-PW input file
#   - Genomic block definition BED file
#
# Replace example phenotype names and file paths as required.
# =====

TRAIT1="Trait1"
TRAIT2="Trait2"

INPUT_FILE="gwaspw.input.${TRAIT1}_${TRAIT2}.tsv"
OUTPUT_PREFIX="${TRAIT1}_${TRAIT2}_post"

# Genomic blocks used by GWAS-PW
BLOCK_FILE="/path/to/genomic_blocks.bed"

/usr/bin/time -v gwas-pw
  -i "${INPUT_FILE}"
  -phenos "${TRAIT1}" "${TRAIT2}"
  -o "${OUTPUT_PREFIX}"
  -bed "${BLOCK_FILE}"
  -k 5000
  -print
  -seed 42
```

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#!/usr/bin/env Rscript

# =====
# Summarise PW-GWAS block-level posterior probabilities
#
# PW-GWAS models:
#
#   P0 = neither trait has an association
#   P1 = association with Trait 1 only
#   P2 = association with Trait 2 only
#   P3 = shared causal variant
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# P4 = two distinct causal variants
#
# The script also:
# - identifies the maximum-a-posteriori (MAP) model
# - reports high-confidence P3/P4 regions
# - optionally removes the MHC and APOE regions
# =====

suppressPackageStartupMessages({
  library(data.table)
})

# -----
# 1. Input settings
# -----

TRAIT1 <- "Trait1"
TRAIT2 <- "Trait2"

stem <- paste0(TRAIT1, "_", TRAIT2)

segfile <- paste0(
  stem,
  "_post_DECsorted.segbfs.gz"
)

cat(
  "Reading PW-GWAS SEGBFS file:",
  segfile,
  "\n"
)

# -----
# 2. Load PW-GWAS results
# -----

seg <- fread(
  segfile,
  sep = " ",
  header = TRUE,
  fill = TRUE
)

# -----
# 3. Construct block-level summary

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# -----

block_summary <- seg[, .(

  block = chunk,
  chr,
  st,
  sp,

  # Posterior probability that only Trait 1
  # contains an association
  P1 = PPA_1,

  # Posterior probability that only Trait 2
  # contains an association
  P2 = PPA_2,

  # Posterior probability that both traits
  # share a causal variant
  P3 = PPA_3,

  # Posterior probability that both traits
  # are associated through distinct causal variants
  P4 = PPA_4
)]

# P0 = posterior probability that neither phenotype
# contains an association in the region

block_summary[, P0 := pmax(
  0,
  1 - (P1 + P2 + P3 + P4)
)]

# -----
# 4. Determine MAP model for each region
# -----

posterior_columns <- c(
  "P0",
  "P1",
  "P2",
  "P3",
  "P4"
)

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block_summary[, MAP_model :=
  max.col(
    .SD,
    ties.method = "first"
  ) - 1L,
  .SDcols = posterior_columns
]

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Highest posterior probability within each block

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block_summary[, MAP_prob :=
  do.call(
    pmax,
    c(
      .SD,
      list(na.rm = TRUE)
    )
  ),
  .SDcols = posterior_columns
]

```

Reorder output columns

```

setcolorder(
  block_summary,
  c(
    "chr",
    "st",
    "sp",
    "block",
    "P0",
    "P1",
    "P2",
    "P3",
    "P4",
    "MAP_model",
    "MAP_prob"
  )
)

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# -----
# 5. Save complete block summary
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output_file <- paste0(

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    stem,
    "_block_summary_DECsorted.tsv"
)

fwrite(
  block_summary,
  output_file,
  sep = "\t"
)

cat(
  "Saved:",
  output_file,
  "\n"
)

# -----
# 6. QC summaries
# -----

cat(
  "\n=== Top blocks by max(P3, P4) ===\n"
)

print(
  head(
    block_summary[
      order(
        -pmax(P3, P4)
      )
    ],
    15
  )
)

cat(
  "\n=== MAP model counts ===\n"
)

print(
  block_summary[
    ,
    .N,
    by = MAP_model
  ][
    order(MAP_model)
  ]
)

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    ]
  )

# -----
# 7. High-confidence regions
# -----

POSTERIOR_THRESHOLD <- 0.90

cat(
  "\n=== High-confidence PW-GWAS regions ===\n"
)

cat(
  "P3 (shared) >=",
  POSTERIOR_THRESHOLD,
  ":",
  nrow(
    block_summary[
      P3 >= POSTERIOR_THRESHOLD
    ]
  ),
  "\n"
)

cat(
  "P4 (distinct) >=",
  POSTERIOR_THRESHOLD,
  ":",
  nrow(
    block_summary[
      P4 >= POSTERIOR_THRESHOLD
    ]
  ),
  "\n"
)

# -----
# 8. Exclude MHC and APOE regions
# -----

# MHC region
MHC_chr    <- 6L
MHC_start  <- 25e6
MHC_end    <- 35e6

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# APOE region
APOE_chr    <- 19L
APOE_start  <- 44.4e6
APOE_end    <- 45.9e6

block_summary_filtered <- block_summary[

  !(chr == MHC_chr &
    st < MHC_end &
    sp > MHC_start)

  &

  !(chr == APOE_chr &
    st < APOE_end &
    sp > APOE_start)
]

# -----
# 9. Save MHC/APOE-excluded results
# -----

filtered_output <- paste0(
  stem,
  "_block_summary_noMHCnoAPOE.tsv"
)

fwrite(
  block_summary_filtered,
  filtered_output,
  sep = "\t"
)

cat(
  "Saved:",
  filtered_output,
  "\n"
)

# -----
# 10. High-confidence regions outside MHC/APOE
# -----

cat(
  "\n=== High-confidence regions outside MHC/APOE ===\n"
)

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)

cat(
  "Shared P3 >=",
  POSTERIOR_THRESHOLD,
  ":",
  nrow(
    block_summary_filtered[
      P3 >= POSTERIOR_THRESHOLD
    ]
  ),
  "\n"
)

cat(
  "Distinct P4 >=",
  POSTERIOR_THRESHOLD,
  ":",
  nrow(
    block_summary_filtered[
      P4 >= POSTERIOR_THRESHOLD
    ]
  ),
  "\n"
)

cat(
  "\nPW-GWAS block summarisation complete.\n"
)

```